

Student Career Success Analysis

Data Analytics Project

1. Problem Statement

Identifying the key drivers of student placement success using academic, skill, and experience data to enable data-driven career decision-making.

2. Dataset

The dataset contains 50,000 student records. It includes academic performance (CGPA, attendance, study hours), technical skills (programming skill, projects, certifications, hackathons), professional experience (internships, leadership experience, GitHub/LinkedIn presence), soft skills (communication, teamwork, problem solving, English proficiency), and placement outcomes (placement status, company tier, career field, starting salary).

3. Steps Used for Analysis

- 1 Collect and load the CSV dataset.
- 2 Check the data for missing values, duplicates, and incorrect/inconsistent values.
- 3 Validate numeric ranges and logical consistency between placement-related columns.
- 4 Detect and cap outliers (Study Hours, CGPA, Starting Salary).
- 5 Group and summarize the data using totals, counts, percentages, and averages.
- 6 Create charts to compare the results.
- 7 Use the results to make career-readiness and placement-strategy decisions.

4. Live Analysis

What is the overall placement rate?

78.08% of students were placed.

Which major has the highest placement rate?

Artificial Intelligence had the highest placement rate at 86.62%, followed closely by Software Engineering at 86.22%.

Which major has the lowest placement rate?

Business Analytics had the lowest placement rate at 59.05%, followed by Electrical Engineering at 60.20%.

Which career field do most placed students enter?

Software Engineer was the most common career field (6,544 students), followed by Data Scientist (3,845) and AI Engineer (3,792).

What is the most common placement mode?

Campus Placement was the most common route, accounting for 19,619 of all placements.

Does having a GitHub profile improve placement chances?

Yes. Students with a GitHub profile had a placement rate of 80.13%, compared to 69.14% for those without one.

How many internships are typically needed to get placed?

Placement rate rises sharply with internship count: 6.78% with 0 internships, 40.73% with 2, and 91.87% with 5 internships.

Do placed students have a higher CGPA than students who are not placed?

Yes. Placed students had an average CGPA of 3.15, compared to 2.96 for students who were not placed.

Do placed students have a higher Employability Score?

Yes. Placed students averaged an Employability Score of 220.71, compared to 190.06 for students not placed.

Which company tier pays the highest starting salary?

Tier 1 companies offered the highest average starting salary at \$90,011, compared to \$57,500 for Tier 2 and \$46,958 for Tier 3.

Which major earns the highest average starting salary among placed students?

Artificial Intelligence graduates earned the highest average starting salary at \$89,536, closely followed by Software Engineering at \$89,360.

5. Career & Institutional Decisions

Placement cells should prioritize internship placement support, since internship count shows the strongest relationship with placement outcomes. Students in lower-placement majors (Business Analytics, Electrical Engineering) should be given targeted upskilling support, such as guided certification tracks and project mentorship. Building a visible GitHub/LinkedIn presence should be actively encouraged early in a student's academic journey, since it correlates with a meaningfully higher placement rate. Career counseling should also highlight Tier 1 recruitment pathways, given the significant salary gap compared to Tier 2 and Tier 3 outcomes.

6. Conclusion

This analysis shows how student academic, skill, and experience data can be converted into actionable insights that help students, faculty, and placement cells understand what drives career readiness and make better, evidence-based decisions to improve placement outcomes.